STOCK MARKET DATA ANALYSIS

End-to-End Data Analysis, SQL Processing & Interactive Business Intelligence

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Data Analytics

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1 Project Overview

1.1 Introduction

Stock market data contains valuable information about price movements, trading activity, company performance, and market behavior. Analyzing this data systematically helps in understanding historical price patterns, comparing companies, examining trading activity, and deriving meaningful insights from structured financial data.

This project focuses on the end-to-end analysis of stock market data using Excel, MySQL, Power Query, and Microsoft Power BI. The project covers the complete analytical workflow, beginning with raw data preparation and cleaning, followed by analytical calculations and exploratory analysis in Excel. The prepared data is then processed and analyzed using MySQL and SQL before being connected to Power BI for further transformation, data modeling, DAX calculations, KPI development, and interactive dashboard creation.

During the project, additional analytical fields were created from the original dataset to support further analysis. These fields include Daily Return %, 5-Day Moving Average, Profit_Value, and Profit/Loss classification.

The project demonstrates how a raw stock market dataset can be transformed into structured analytical information by integrating spreadsheet analysis, relational database processing, and business intelligence techniques.

1.2 Business and Analytical Context

The original stock market dataset contains information such as trading date, company, opening price, highest price, lowest price, closing price, adjusted closing price, and trading volume.

Additional analytical fields were calculated during the project to enhance the dataset and support the required analysis. Daily Return % was calculated to measure the percentage change in the stock price between the trading periods. The 5-Day Moving Average was calculated to examine the short-term price movement. Profit_Value was created to represent the numerical profit or loss, and the Profit/Loss field was created to classify records into Profit and Loss categories.

The analysis focuses on examining price behavior, trading activity, company-level performance, and profit/loss patterns. Excel is used for initial data preparation, cleaning, calculations, and exploratory analysis. MySQL is used for structured data storage, processing, and SQL-based analysis. Power Query is used for further data transformation, while Power BI is used for data modeling, normalization, DAX-based calculations, KPI development, and interactive visualization.

The final analysis provides multiple views of the dataset through interactive dashboards, including overall market-level information, price and company analysis, profit and loss analysis, and company-specific drill-through analysis.

1.3 Project Objectives

The primary objectives of the project are:

- To understand the structure and characteristics of the original stock market dataset.
- To clean and prepare the dataset by validating data types, dates, numerical fields, duplicate records, and missing values.

- To create analytical fields such as Daily Return %, 5-Day Moving Average, Profit_Value, and Profit/Loss classification to support further analysis.
- To perform exploratory analysis using Excel and create meaningful calculations and pivot tables.
- To store and process the prepared dataset using a MySQL relational database.
- To perform SQL-based analysis for extracting relevant company, price, volume, and profit/loss information.
- To transform and prepare the data using Power Query for business intelligence analysis.
- To develop a structured Power BI data model using a fact table and supporting dimension tables.
- To create DAX measures and KPIs for analyzing trading activity, closing prices, profit days, loss days, and related metrics.
- To develop interactive Power BI dashboards for overall, company-level, price, volume, and profit/loss analysis.
- To implement company-level drill-through analysis for detailed examination of individual companies.
- To identify data-supported, meaningful business insights from the analyzed stock market data.
- To validate important analytical results across Excel, MySQL, and Power BI to ensure consistency.

2 Dataset Structure

2.1 Dataset Overview

The stock market dataset used in this project contains 300 records and 12 columns in its final analytical form. Each record represents a stock market observation containing information related to the trading date, company, stock prices, trading volume, daily return, moving average, and profit or loss.

The original dataset contained 8 columns. During the data preparation and analysis stages, 4 additional analytical columns were created. Therefore, the final dataset used for analysis contains 12 columns.

Attribute	**Value**
Number of Records	300
Original Columns	8
Created Columns	4
Final Columns	12
Dataset Type	Stock Market Data
Primary Analysis	Price, Volume and Profit/Loss

Table 1: Dataset Overview

2.2 Dataset Columns

The final dataset used for the analysis contains the following fields:

Column	**Data Type**	**Description**
Date	Date	Trading date of the stock market record.
Company	Text	Name of the company.
Open	Numeric	Opening price of the stock.
High	Numeric	Highest price recorded during the trading period.
Low	Numeric	Lowest price recorded during the trading period.
Close	Numeric	Closing price of the stock.
Adj Close	Numeric	Adjusted closing price of the stock.
Volume	Numeric	Number of shares traded.
Daily Return %	Numeric	Calculated percentage change in the closing price.
5-Day Moving Average	Numeric	Calculated moving average of the closing price over five trading days.
Profit_Value	Numeric	Calculated numerical profit or loss value.
Profit/Loss	Text	Created classification indicating Profit or Loss.

Table 2: Data Dictionary

2.3 Original and Created Fields

The dataset used in the project consists of original fields from the source dataset and additional fields created during the analysis.

2.3.1 Original Fields

The original dataset contained the following eight fields:

- **Date, Company, Open, High, Low, Close, Adj Close, Volume**

These fields represent the basic stock market information available in the source dataset.

2.3.2 Created Analytical Fields

Four additional fields were created during the project to support further analysis:

- **Daily Return %** - calculated to measure the percentage change in the closing price.
- **5-Day Moving Average** - calculated to examine short-term price movement by using the average closing price over five trading days.
- **Profit_Value** - created to represent the numerical profit or loss value.
- **Profit/Loss** - created to classify each record as either Profit or Loss.

The resulting dataset contains 12 columns and was used for subsequent Excel analysis, MySQL processing, SQL analysis, Power Query transformation, Power BI data modeling, and dashboard development.

2.4 Original Dataset

The original dataset used as the starting point for the project is shown below.

	A	B	C	D	E	F	G	H	I
1	Date	Company	Open	High	Low	Close	Adj Close	Volume	
2	2025-03-07	TCS	2892.97	2908.95	2878.99	2893.28	2891.9	114835	
3	2025-03-07	INFY	1266.67	1271.89	1259.93	1262.37	1260.55	192803	
4	2025-03-07	WIPRO	1949.49	1957.29	1940.21	1946.31	1945	323996	
5	2025-03-07	HDFC	688.6	703.98	670.86	692.82	691.82	466456	
6	2025-03-07	RELIANCE	2354.74	2374.55	2349.14	2354.52	2355.99	133075	
7	2025-03-10	TCS	2890.79	2906.41	2885.48	2901	2902.88	327903	
8	2025-03-10	INFY	1251.63	1258.1	1239.26	1258.04	1257.93	165944	
9	2025-03-10	WIPRO	1930.97	1950.71	1924.29	1940.92	1940.51	156769	
10	2025-03-10	HDFC	667.73	687.55	648.57	667.77	665.93	474865	
11	2025-03-10	RELIANCE	2351.38	2367.26	2339.08	2363.18	2364.67	151542	
12	2025-03-11	TCS	2883.31	2891.04	2875.56	2886.01	2885.23	284736	
13	2025-03-11	INFY	1251.47	1262.45	1237.23	1243.6	1244.14	250957	
14	2025-03-11	WIPRO	1950.23	1957.79	1935.2	1950.27	1951.99	204247	
15	2025-03-11	HDFC	687.3	695.01	673.78	685.43	687.09	342063	
16	2025-03-11	RELIANCE	2365.27	2374.42	2351.38	2367.32	2368.97	305545	
17	2025-03-12	TCS	2861.87	2871.24	2847.69	2864.22	2862.78	488073	
18	2025-03-12	INFY	1252.72	1267.11	1240.17	1251.31	1252.74	474764	
19	2025-03-12	WIPRO	1976.43	1985.63	1959.89	1980.86	1979.61	340506	
20	2025-03-12	HDFC	687.47	707.04	668.31	686.61	686.51	316105	
21	2025-03-12	RELIANCE	2372.33	2386.8	2356.33	2365.3	2363.83	227627	
22	2025-03-13	TCS	2878.92	2890.76	2862.14	2869.39	2868.19	390916	

Figure 1: Original Stock Market Dataset

3 Project Workflow

The project follows an end-to-end data analytics workflow that transforms the raw stock market dataset into structured analytical information and interactive business intelligence dashboards. The workflow integrates Excel, MySQL, Power Query, data modeling, DAX, and Power BI.

The process begins with the raw stock market dataset and proceeds through data preparation and cleaning, analytical calculations, MySQL database processing and SQL analysis, Power Query transformation, data modeling and normalization, DAX measure and KPI development, and interactive Power BI dashboard development. The final stage involves deriving business insights and validating the analytical results.

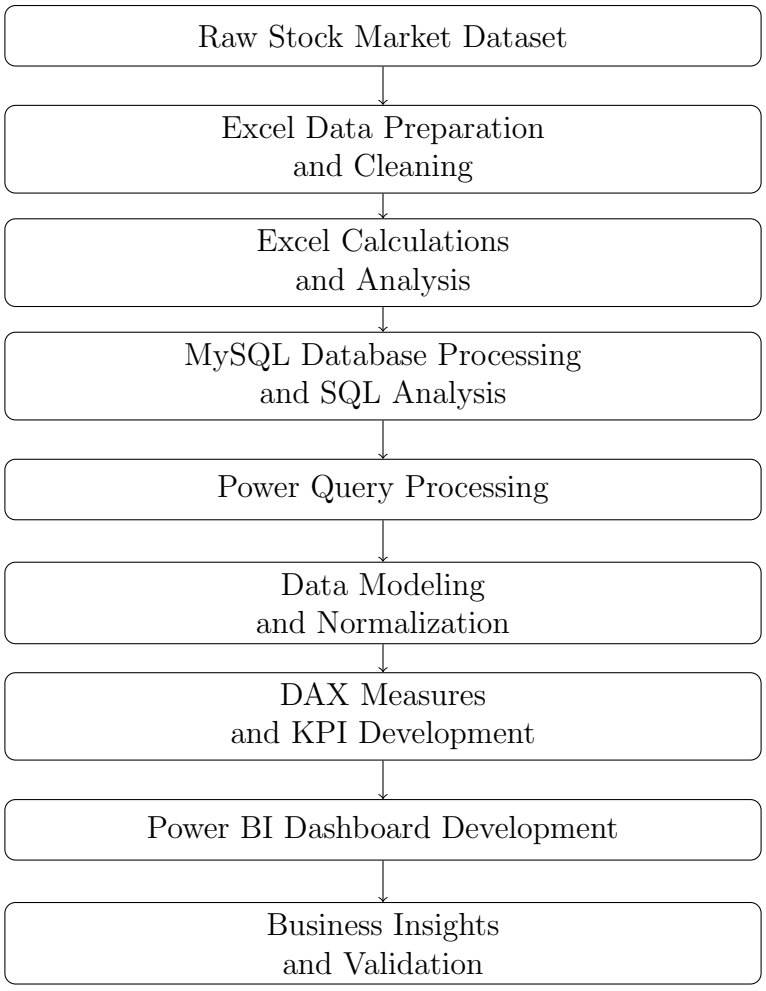


Figure 2: End-to-End Project Workflow

4 Excel – Data Preparation, Cleaning and Analysis

4.1 Data Preparation

The stock market dataset was examined in Excel to understand its structure and prepare the data for analysis. The preparation stage focused on reviewing the available fields, record count, data types, date values, and numerical columns.

The preparation process included:

- Reviewing the available columns.
- Checking the number of records.
- Examining the data types of the columns.
- Reviewing the date values.
- Checking numerical fields.
- Reviewing the analytical fields added to the dataset.

4.2 Data Cleaning

Data cleaning and validation checks were performed to improve the quality and consistency of the dataset before further analysis.

4.2.1 Duplicate Check

The dataset was checked for duplicate records to ensure that repeated records did not affect calculations, aggregations, or subsequent analysis.

4.2.2 Null Value Check

The relevant columns were checked for missing or null values. This check was performed to ensure that missing values did not affect calculations and analytical results.

4.2.3 Date Standardization

The date field was reviewed and standardized to ensure consistent date representation and proper interpretation during subsequent MySQL and Power BI processing.

4.2.4 Numeric and Data-Type Validation

The numerical fields, including Open, High, Low, Close, Adj Close, Volume, Daily Return %, 5-Day Moving Average, and Profit_Value, were checked to ensure that the values were stored in appropriate numerical formats.

4.3 Calculated Fields

Additional analytical fields were created to support the analysis of stock price movement and profit/loss behavior.

4.3.1 Daily Return %

Daily Return % was created to measure the percentage movement in the stock price for each record. This field provides an additional measure for examining changes in stock price over the trading period.

4.3.2 Profit/Loss Classification

The numerical Profit_Value field was used to classify records into two categories:

- Profit
- Loss

The resulting Profit/Loss field was later used for filtering, aggregation, and profit/loss analysis in Power BI.

4.3.3 5-Day Moving Average

A 5-Day Moving Average was created to examine short-term price movement by calculating the average closing price over five trading days. This field helps provide a smoother representation of price movement compared with individual closing prices.

4.4 Pivot Table Analysis

Pivot Tables were created in Excel to summarize the stock market dataset from different analytical perspectives. The main analyses included company-wise price, trading volume, and profit/loss information.

4.4.1 Company-wise Closing Price Analysis

The Pivot Table was used to summarize closing price information for each company.

Company	Average of Close
HDFC	682.7615
INFY	1254.441
RELIANCE	2367.928167
TCS	2889.199
WIPRO	1954.354833
Grand Total	**1829.7369**

4.4.2 Company-wise Trading Volume Analysis

The Pivot Table was used to compare trading volume across companies.

Company	Sum of Volume
HDFC	19271942
INFY	18604961
RELIANCE	16102795
TCS	18391257
WIPRO	18356148
Grand Total	**90727103**

4.4.3 Profit/Loss Analysis

The Pivot Table was used to summarize the distribution of Profit and Loss records.

Profit/Loss	Count of Date
Loss	146
Profit	154
Grand Total	**300**

4.5 Excel Charts

Charts were created to visually represent the results obtained from the Excel analysis. These visualizations helped in examining company performance, trading activity, and profit/loss behavior.

4.5.1 Company-wise Analysis Chart

The chart provides a visual comparison of the selected company-wise measure.



Figure 3: Company-wise Analysis

4.5.2 Trading Volume Chart

The chart provides a visual representation of trading volume across the analyzed companies.

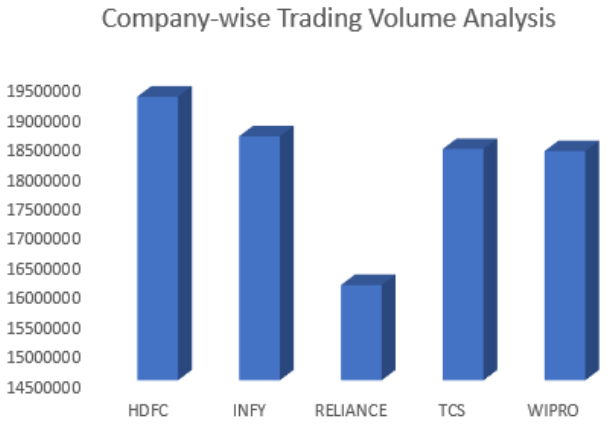


Figure 4: Trading Volume Analysis

4.5.3 Profit/Loss Chart

The chart presents the distribution of Profit and Loss records in the dataset.

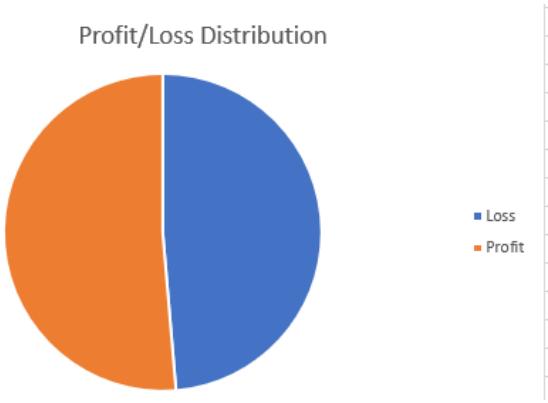


Figure 5: Profit/Loss Analysis

4.6 Excel Analysis Results

The calculations, Pivot Tables, and charts provided an initial analytical view of the stock market dataset. The results were used as a reference for the subsequent MySQL and Power BI analysis.

Important analytical results were cross-checked during the later stages of the project to maintain consistency across the different tools.

5 MySQL – Database Processing and SQL Analysis

5.1 Database Creation

A MySQL database was created to store and process the prepared stock market dataset. The database provided a structured relational environment for storing the data and performing SQL-based analysis.

```
create database stock_market;
use stock_market;
```

5.2 Table Structure

A main stock market table was created to store the 12-column analytical dataset.

```
create table stock_data(
    Date date,
    Company varchar(100),
    Open decimal(10,2),
    High decimal(10,2),
    Low decimal(10,2),
    Close decimal(10,2),
    Adj_Close decimal(10,2),
    Volume int,
    Daily_Return_Percent decimal(10,3),
    Moving_Average_5_Day decimal(10,3),
    Profit_Value decimal(10,2),
    Profit_Loss varchar(100)
);
```

The table structure was designed to accommodate the original stock market fields along with the additional analytical fields created during the project.

5.3 Data Import

The prepared dataset was imported into MySQL after checking the required column types and date format.

During the initial import, the date format of the source data required conversion to a format compatible with the MySQL `DATE` data type. After correcting the date representation, the dataset was successfully imported into the MySQL table.

The imported data was then checked to confirm that the records and fields were loaded correctly.

5.4 Database Processing

After importing the dataset, basic database checks were performed to verify the stored data.

The processing and verification included:

- Checking the total number of records.

- Verifying the table columns.
- Checking the stored data types.
- Verifying the date values.
- Inspecting sample records.

5.5 SQL Analysis

SQL queries were used to extract, summarize, and analyze information from the stock market table. The analysis focused on company-level performance, closing prices, trading volume, and profit/loss behavior.

5.5.1 Top 5 Closing Prices

The following query was used to identify the five records with the highest closing prices.

```
select Company, Date, Close from stock_data
order by Close
desc limit 5;
```

Company	Date	Close
TCS	2025-05-23	2920.44
TCS	2025-04-01	2920.13
TCS	2025-05-16	2913.52
TCS	2025-04-17	2912.36
TCS	2025-04-30	2912.24

Table 3: SQL Result for Top 5 Closing Prices

5.5.2 Average Volume by Company

The average trading volume for each company was calculated using the following query.

```
select Company, avg(Volume) as Average_Volume
from stock_data
group by Company;
```

Company	Average_Volume
TCS	306520.9500
INFY	310082.6833
WIPRO	305935.8000
HDFC	321199.0333
RELIANCE	268379.9167

Table 4: SQL Result for Average Volume by Company

5.5.3 Profit-Day Analysis

The number of profitable trading records for each company was obtained using the following query.

```
select Company, count(*) as Profit_Days
from stock_data
where close > open
group by Company;
```

Company	Profit_Days
TCS	32
HDFC	29
INFY	29
WIPRO	25
RELIANCE	39

Table 5: SQL Result for Profit-Day Analysis

5.6 SQL Query Results

The SQL query results provided an additional analytical view of the stock market dataset. These results were used to examine company-level price and volume information and profit/loss behavior.

The SQL results were also used as a reference during the subsequent Power BI analysis and validation process.

6 Power BI – Data Processing and Dashboard Development

6.1 Power BI Data Connection

Power BI was connected to the MySQL database to obtain the processed stock market dataset. The successfully imported MySQL table was loaded into Power BI for further transformation, modeling, analysis, and visualization.

The imported data was reviewed before proceeding with the Power Query transformation and data modeling stages.

6.2 Power Query Processing

Power Query was used to transform and prepare the data before creating the final Power BI data model.

The main processing activities included:

1. Loading the required data from MySQL.
2. Reviewing and validating column names.
3. Checking data types.
4. Validating the date field.
5. Reviewing numerical fields.
6. Preparing the fact table.
7. Preparing the required dimension tables.
8. Removing unnecessary fields where applicable.

The transformed data was then used to create the structured Power BI data model.

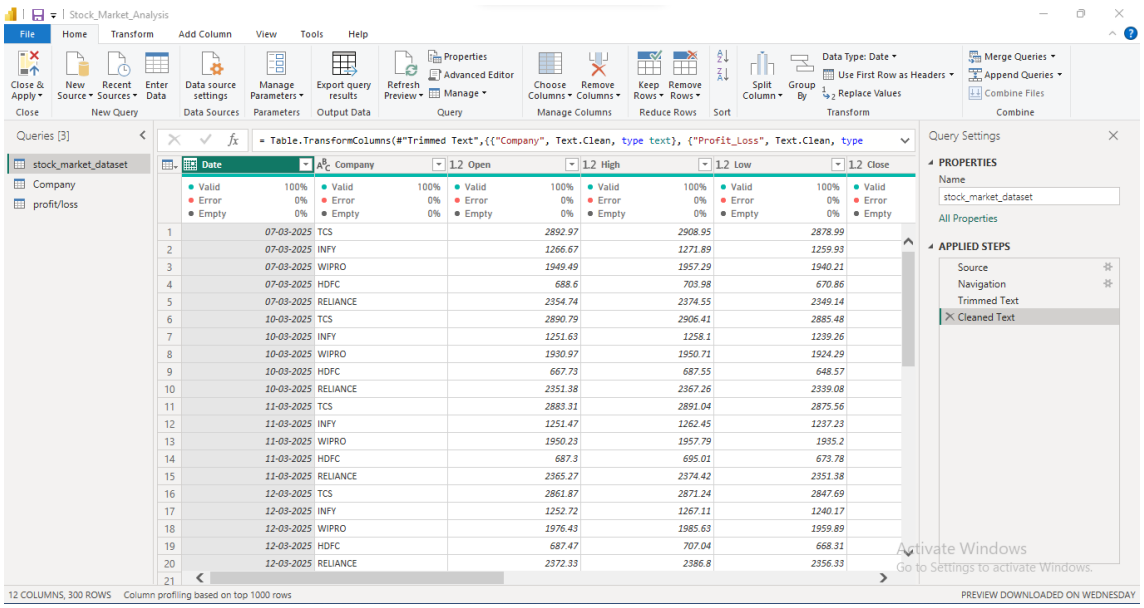


Figure 6: Power Query Data Transformation

6.3 Data Modeling and Normalization

A structured data model was created in Power BI to organize the stock market data into a fact table and supporting dimension tables. This structure separates the main analytical records from descriptive information and supports efficient filtering and aggregation.

The Power BI model consists of the following tables:

- stock_market_dataset
- date_table
- company
- profit/loss

6.3.1 stock_market_dataset

The Fact stock_market_dataset table contains the main stock market records, including price, volume, return, moving average, and profit/loss information. It acts as the central table for the analytical model.

6.3.2 date_table

The date_table provides a dedicated date dimension for date-based filtering and time-related analysis in Power BI.

6.3.3 company

The Dimension company table contains the unique company values used for company-level filtering, comparison, and analysis.

6.3.4 profit/loss

The Dimension profit/loss table contains the Profit and Loss classification values used for filtering and profit/loss analysis.

6.3.5 Relationships

Relationships were created between the dimension tables and the Fact stock_market_dataset table to support filtering and aggregation across the Power BI report.

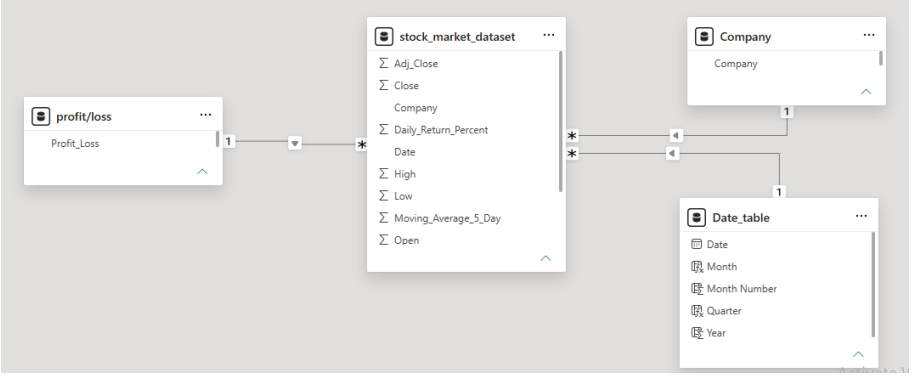


Figure 7: Power BI Data Model

6.4 DAX Measures

DAX measures were created to calculate the key analytical values required for the Power BI dashboards.

6.4.1 Total Volume

```
Total Volume = SUM(stock_market_dataset[Volume])
```

This measure calculates the total trading volume.

6.4.2 Average Closing Price

```
Average Closing Price = AVERAGE(stock_market_dataset[Close])
```

This measure calculates the average closing price.

6.4.3 Total Profit

```
Total Profit = SUM(stock_market_dataset[Profit_Value])
```

This measure calculates the total numerical profit or loss value available in the dataset.

6.4.4 Profit Days

```
Profit Days = CALCULATE(COUNTROWS(stock_market_dataset),
              stock_market_dataset[Profit_Loss] = "Profit" )
```

This measure counts the records classified as Profit.

6.4.5 Loss Days

```
Loss Days = CALCULATE(COUNTROWS(stock_market_dataset),
            stock_market_dataset[Profit_Loss] = "Loss" )
```

This measure counts the records classified as Loss.

6.4.6 Trading Days

```
Trading Days = COUNTROWS(stock_market_dataset)
```

This measure counts the trading days available in the dataset.

6.4.7 Profit Percentage

```
Profit % = DIVIDE([Profit Days],[Profit Days]+[Loss Days],0)
```

This measure calculates the percentage value used to represent profitable trading records in the dashboard.

6.4.8 Loss Percentage

```
Loss % = DIVIDE([Loss Days],[Profit Days]+[Loss Days],0)
```

This measure calculates the percentage value used to represent loss-making trading records in the dashboard.

6.5 KPI Development

The DAX measures were used to create the primary KPIs displayed in the Power BI dashboards.

KPI	Purpose
Total Volume	Measures overall trading activity.
Average Closing Price	Represents the average closing price.
Total Profit	Represents the total numerical profit or loss value.
Profit Days	Counts records classified as Profit.
Loss Days	Counts records classified as Loss.
Profit %	Represents the calculated proportion of profitable records.
Loss %	Represents the calculated proportion of loss-making records.
Trading Days	Counts trading days.

Table 6: Power BI KPIs

6.6 Dashboard Development

The Power BI report was developed as a collection of interactive dashboard pages. Each page was designed for a specific analytical purpose while allowing users to interact with the data through slicers, filters, and drill-through functionality.

6.6.1 Executive Dashboard

The Executive Dashboard provides a high-level summary of the stock market dataset. The page includes:

- KPI cards.
- Price-related visualizations.
- Trading volume analysis.
- Company comparison.
- Interactive slicers and filters.

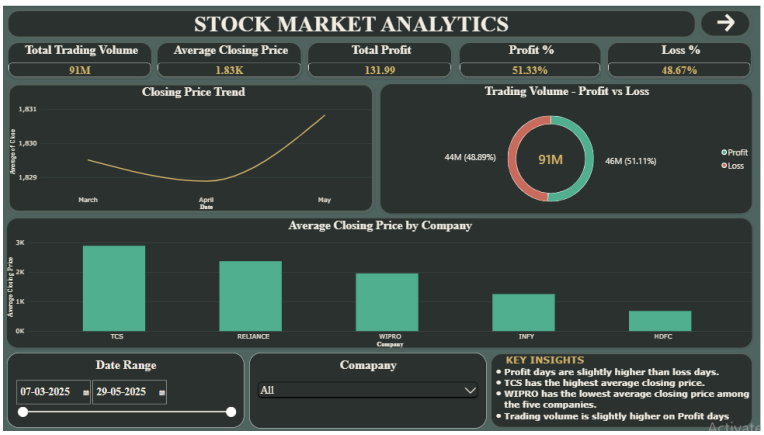


Figure 8: Power BI Executive Dashboard

6.6.2 Price and Company Analysis

The Price and Company Analysis page focuses on stock price behavior and comparison between companies.

The page contains price-related visualizations, company comparisons, and interactive filtering options.

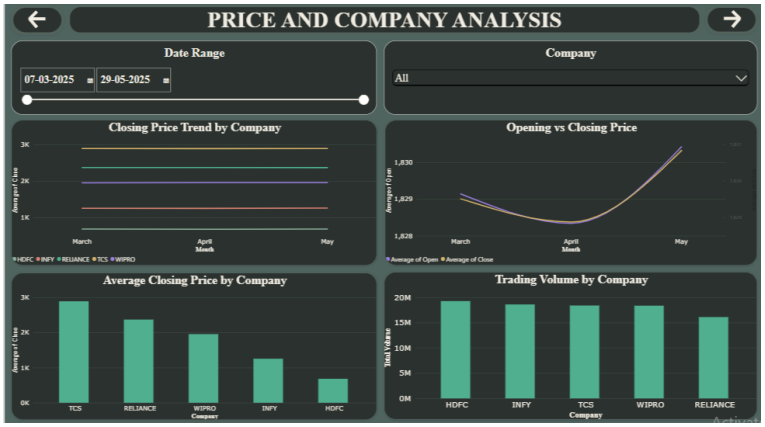


Figure 9: Price and Company Analysis Dashboard

6.6.3 Profit and Loss Analysis

The Profit and Loss Analysis page focuses on profitable and loss-making records within the dataset.

The page uses KPIs, charts, and company-level analysis to examine profit/loss behavior.

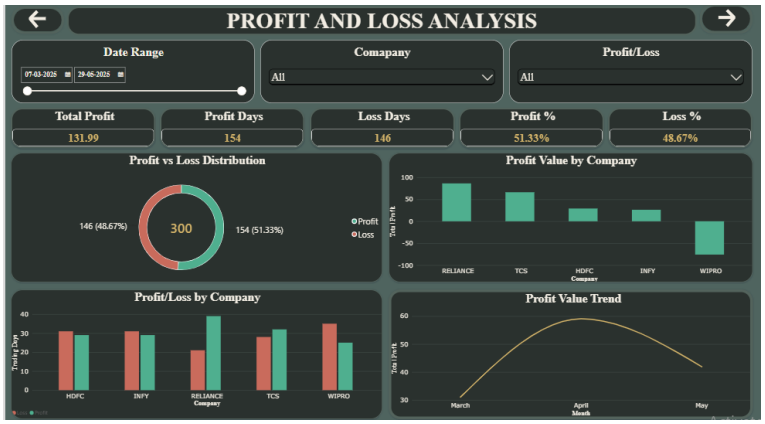


Figure 10: Profit and Loss Analysis Dashboard

6.6.4 Company Details and Drill-through

The Company Details page provides detailed analysis for an individual company selected through Power BI drill-through functionality.

The page provides company-specific information and allows the user to examine the selected company in greater detail.

The analysis includes:

- Company-specific KPIs.

- Opening and closing price analysis.
- Profit/loss information.
- Trading volume analysis.
- Company-specific filtering.
- Drill-through functionality.

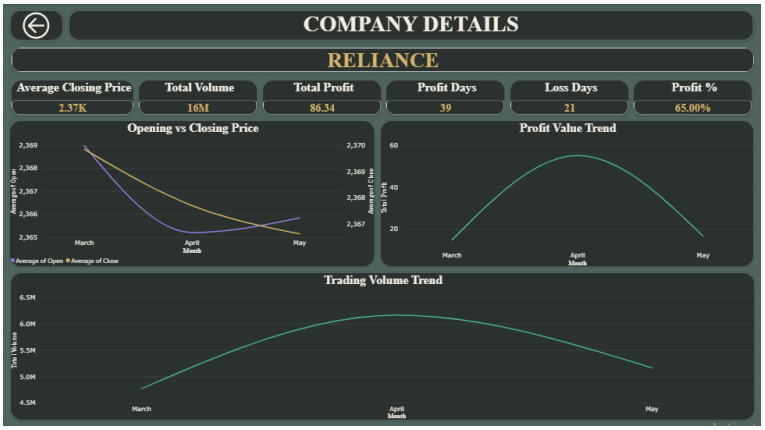


Figure 11: Company Details and Drill-through

7 Business Insights

The following insights were derived from the analysis performed on the stock market dataset using the Power BI dashboards. The analysis provides an overview of company-level price performance, profit and loss distribution, trading activity, and price trends.

7.1 Company-Level Price Analysis

The average closing price varied significantly among the five companies. TCS recorded the highest average closing price at approximately 2.89K, followed by Reliance at approximately 2.37K and Wipro at approximately 1.95K. Infosys recorded an average closing price of approximately 1.25K, while HDFC recorded the lowest average closing price at 682.76.

The results show a clear difference in the price levels of the companies during the analyzed period, with TCS having the highest average closing price and HDFC having the lowest.

7.2 Profit and Loss Distribution

The overall analysis contained 300 records, of which 154 were classified as Profit and 146 were classified as Loss. Profit records represented 51.33% of the total records, while Loss records represented 48.67%.

At the company level, Reliance recorded the highest number of Profit Days with 39, followed by TCS with 32, Infosys and HDFC with 29 each, and Wipro with 25. Reliance also recorded the highest Profit percentage at 65%.

These results indicate that profitable records were slightly higher than loss-making records in the overall dataset.

7.3 Trading Volume Behavior

The total trading volume across the dataset was approximately 91M. Trading volume associated with Profit records was approximately 46M, representing 51.11%, while Loss records accounted for approximately 44M, representing 48.89%.

At the company level, HDFC recorded the highest trading volume at approximately 19M, followed closely by Infosys at approximately 19M. TCS and Wipro recorded approximately 18M each, while Reliance recorded approximately 16M.

The overall trading volume trend increased from March and reached its highest level around April before declining toward May.

7.4 Price Trends

The overall closing price trend showed a decline from March toward April, followed by an increase toward May. The Opening versus Closing Price analysis showed similar movement patterns across the analyzed period.

The company-level trends showed different price behaviors. TCS and Reliance showed a decline toward April followed by an increase in May, while Wipro increased toward April before declining toward May. Infosys and HDFC showed a decline toward April followed by a recovery in May.

The 5-Day Moving Average was used to provide a smoother representation of short-term price movement.

7.5 Other Observations

The company-level profit analysis showed noticeable differences among the five companies. Reliance recorded the highest total profit at 86.34, followed by TCS at 66.17, HDFC at 29.15, and Infosys at 26.24. Wipro recorded a negative total profit of -75.91.

Reliance also recorded 39 Profit Days and 21 Loss Days, resulting in a Profit percentage of 65%. In comparison, TCS recorded 32 Profit Days and 28 Loss Days, resulting in a Profit percentage of 53.33%.

The Profit Value trends varied across companies. Reliance showed a positive increase toward April followed by a decline, while TCS showed a strong increase toward May. HDFC showed a decline toward April followed by a recovery, whereas Wipro remained negative throughout the analyzed period.

Overall, the analysis shows that Reliance demonstrated the strongest profit performance among the five companies, while Wipro recorded the weakest profit result during the analyzed period.

8 Validation and Quality Checks

Validation and quality checks were performed at different stages of the project to ensure that the prepared data, calculations, and analytical results were consistent across Excel, MySQL, and Power BI.

8.1 Excel Validation

The prepared dataset was reviewed in Excel to verify the number of records, data types, date values, numerical fields, duplicate records, and missing values. The calculated fields and Pivot Table results were also checked against the underlying dataset to ensure that the calculations and summaries were generated correctly.

8.2 MySQL Validation

After importing the prepared dataset into MySQL, the table structure and stored records were verified. Record counts, sample records, data types, date values, and SQL query results were checked to ensure that the imported data matched the prepared dataset.

The MySQL data was also checked after resolving the date-format issue encountered during the initial import.

8.3 Power BI Validation

The Power BI model and dashboard calculations were checked against the underlying MySQL data. The KPI values, company-level results, and profit/loss calculations were reviewed using different filters and company selections.

The overall Power BI dashboard showed 300 records, with 154 Profit Days and 146 Loss Days. The corresponding Profit and Loss percentages were 51.33% and 48.67%, respectively. The total trading volume was approximately 91M.

8.4 Cross-Platform Validation

Important analytical results were cross-checked between the Excel, MySQL, and Power BI stages. This helped verify that data transformations, calculations, aggregations, and filtering operations did not introduce inconsistencies during the movement of data between the different tools.

The main validation checks performed during the project are summarized in Table **??**.

Check	Validation Method
Record Count	Compared the number of records between the prepared dataset, MySQL table, and Power BI analysis. The final dataset contained 300 records.
Total Volume	Compared the total trading volume with the corresponding calculations and Power BI KPI result. The overall value was approximately 91M.
Average Closing Price	Checked the average closing price in the analytical calculations and Power BI dashboard. The overall dashboard value was approximately 1.83K.
Profit Days	Verified the number of records classified as Profit. The Power BI dashboard showed 154 Profit Days.
Loss Days	Verified the number of records classified as Loss. The Power BI dashboard showed 146 Loss Days.
Profit and Loss Percentage	Verified the percentage distribution of Profit and Loss records. The dashboard showed 51.33% Profit and 48.67% Loss.
Company Values	Verified company-level results using company filters, SQL analysis, and Power BI drill-through analysis.
Trading Dates	Checked the date values and unique trading dates used for date-based analysis.

Table 7: Validation Checks

9 Project Outcome

The project resulted in the successful development of an end-to-end stock market data analytics solution. The raw stock market dataset was transformed into a structured analytical dataset through data preparation, cleaning, calculation, database processing, and business intelligence analysis. The major outcomes of the project include:

- A cleaned and structured analytical dataset containing 300 records and 12 columns.
- Creation of analytical fields including Daily Return %, 5-Day Moving Average, Profit_Value, and Profit/Loss classification.
- Excel-based data preparation, cleaning, calculations, Pivot Table analysis, and visualization.
- A MySQL database for structured storage and processing of the prepared stock market data.
- SQL-based analysis for examining company-level prices, trading volume, and profit/loss information.
- Power Query transformations for preparing the data for Power BI analysis.
- A structured Power BI data model consisting of a fact table and supporting dimension tables.
- Development of DAX measures and KPIs for price, volume, profit/loss, and trading activity analysis.
- Development of four interactive Power BI dashboard pages covering executive analysis, price and company analysis, profit and loss analysis, and company-level drillthrough analysis.
- Identification of data-supported business insights, including differences in company price levels, trading volume, profit/loss distribution, and company-level profit performance.
- Validation of important analytical results across Excel, MySQL, and Power BI to improve consistency and reliability.

Overall, the project demonstrated the complete process of transforming raw stock market data into structured analytical information and interactive business intelligence dashboards using multiple data analytics tools.

10 Conclusion

This project successfully demonstrated an end-to-end approach to stock market data analytics, transforming a raw dataset into structured analytical information and interactive business intelligence dashboards.

The project integrated Excel, MySQL, Power Query, and Power BI to perform data preparation, cleaning, calculation, database processing, SQL analysis, data transformation, modeling, DAX development, KPI creation, and visualization. Each tool contributed to a different stage of the analytical workflow, resulting in a structured and interactive solution for examining stock prices, trading volume, and profit/loss behavior.

The analysis provided meaningful insights into differences among companies, including price levels, trading activity, profit and loss distribution, and company-level profit performance. The Power BI dashboards further enabled interactive analysis through filtering and company-level drill-through functionality.

Validation across Excel, MySQL, and Power BI helped maintain consistency between the different stages of the project and increased confidence in the analytical results.

Overall, the completed project provided practical experience in data cleaning, exploratory analysis, SQL, relational database processing, Power Query, data modeling, DAX, KPI development, data visualization, and business intelligence reporting. It demonstrates the complete process of converting raw stock market data into useful, data-supported business insights.